

VAPO: Value-Model-Based RL for Advanced Reasoning.4

AIME 2024 avg@32

New SOTA

Base Model

Qwen2.5-32B

Training

No SFT Data

Steps

~5,000

Authors

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Why Value Models? Finer Credit Assignment for Long CoT

Value-Model-Free (GRPO / DAPO)

- Trajectory-level reward assigned uniformly across all tokens
- Simple and effective for basic tasks
- Limited credit assignment for multi-step reasoning
- Errors at any step treated equally

VS

Value-Model-Based (VAPO)

- Token-level credit assignment
- Traces each action's impact on returns
- Lower-variance estimates vs Monte Carlo
- Generalizes across samples → better sample efficiency

Key Insight: Critical for subtle step-level errors that cause catastrophic failures in complex reasoning

Three Challenges Plaguing Value-Model RL in Long-CoT

01

Value Model Bias

Long trajectories and bootstrapped learning create accumulating bias in value estimates

02

Heterogeneous Lengths

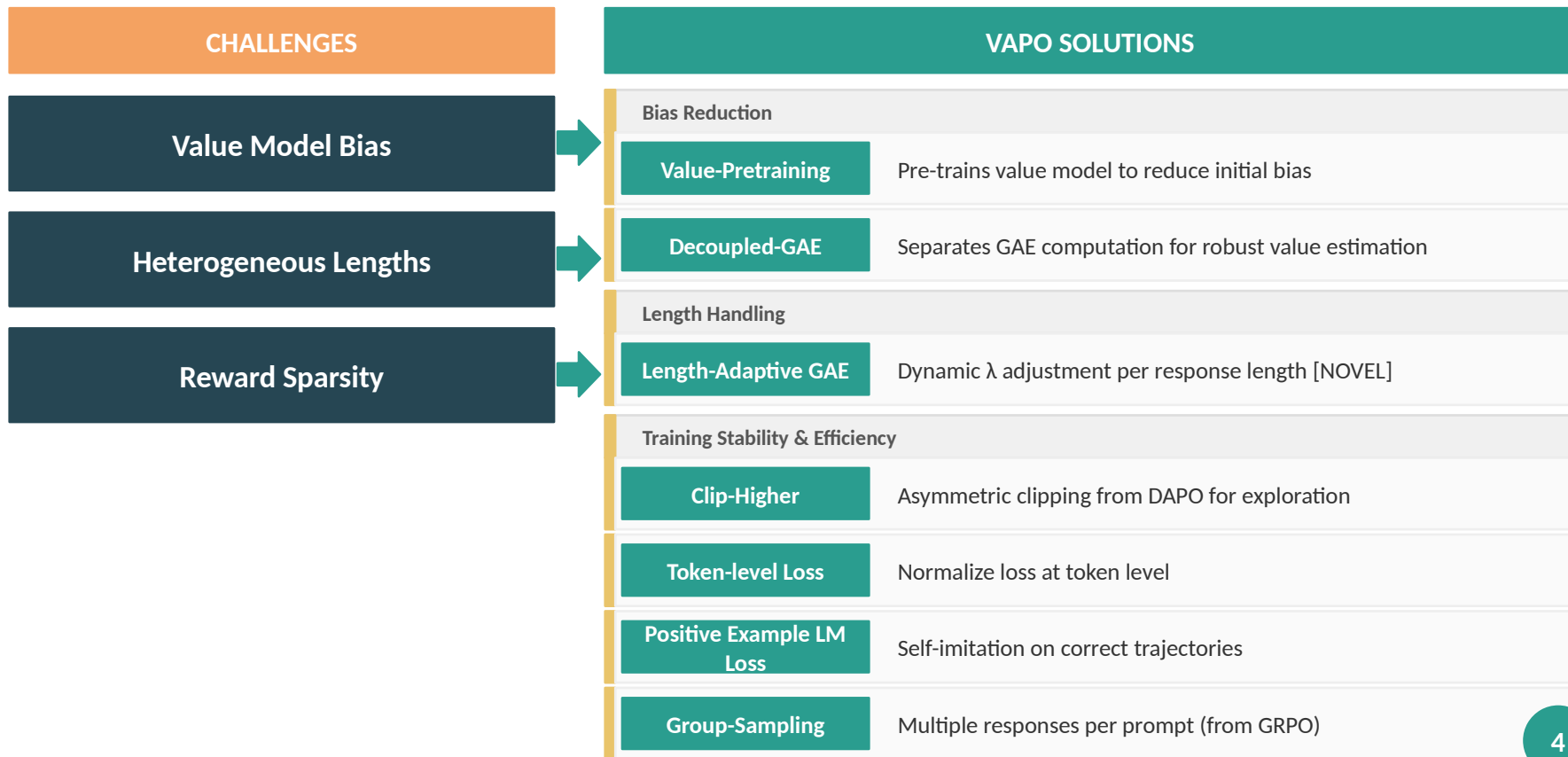
Variable response lengths demand different bias-variance tradeoffs — one λ fits none

03

Reward Sparsity

Only the final <eos> token receives reward; long CoT amplifies this sparsity problem

VAPo Framework: Integrated Solutions for Value-Model RL



Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Standard GAE uses fixed λ — but short vs long CoT have opposite bias-variance needs

SHORT Responses

$\lambda = \text{LOW}$

Prioritizes:

Lower Variance

Fewer steps → bootstrapped estimate is reliable → accept lower bias tolerance



low λ

LONG Responses

$\lambda = \text{HIGH}$

Prioritizes:

Lower Bias

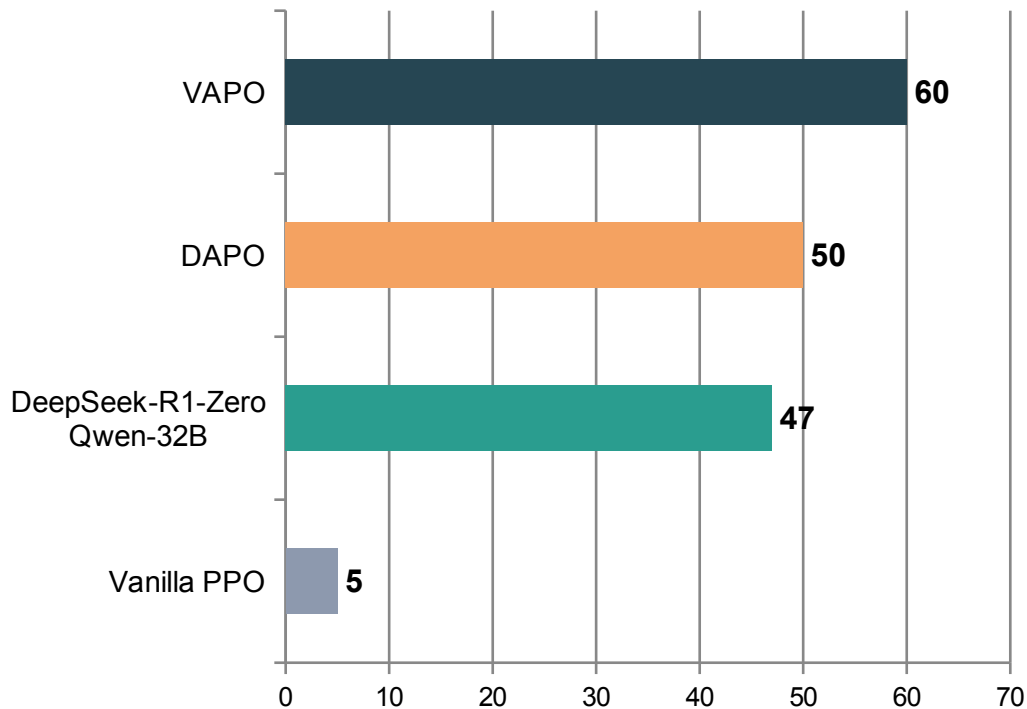
Many steps → bootstrapped estimate drifts → need higher λ to reduce bias accumulation



high λ

Result: +15 points on AIME 2024 — removing Length-Adaptive GAE drops score from 60 → 45

VAPO Scores 60 on AIME 2024 — 10+ Points Above All Baselines



60

AIME 2024 avg@32

+10 pts

vs DAPO (50)

+13 pts

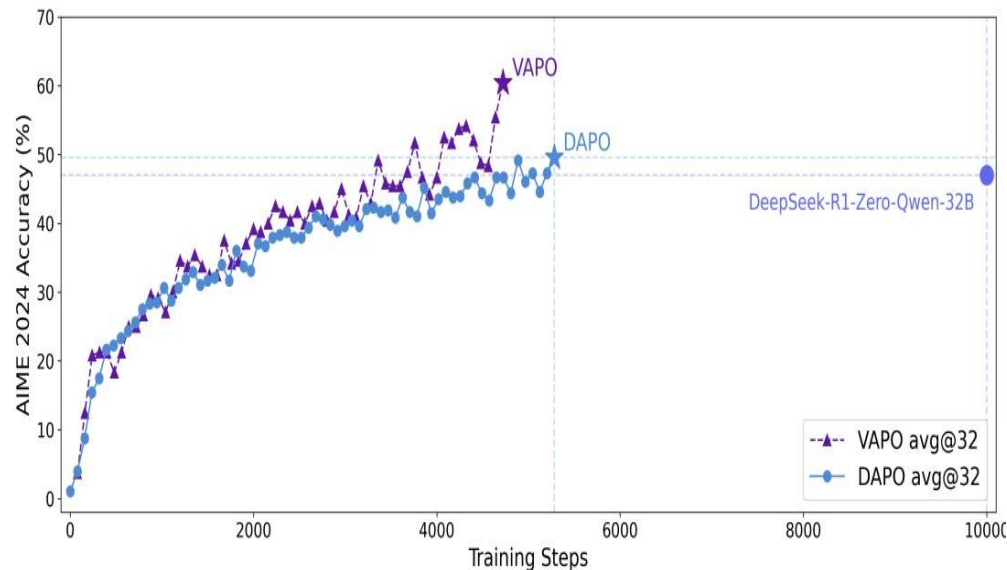
vs DeepSeek-R1-Zero (47)

Base: Qwen2.5-32B, no SFT data

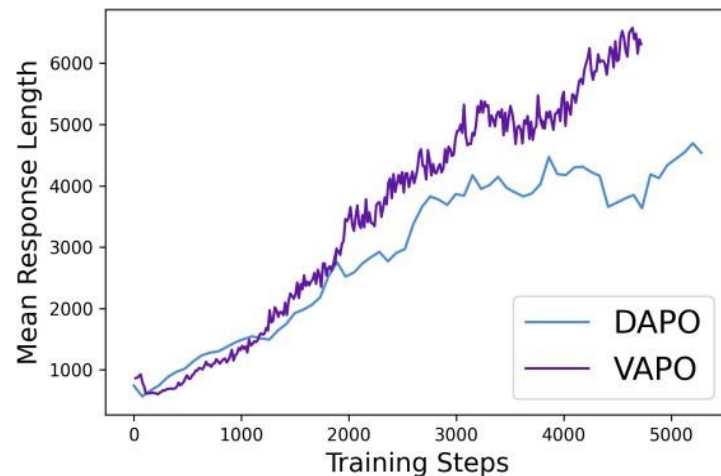
~5,000 training steps

VAPO Reaches SOTA in ~5K Steps — Stable Across All Runs

Training Accuracy (AIME 2024)



Mean Response Length



VAPO generates ~6,000 tokens vs DAPO's ~4,500

~5,000
Steps to SOTA

0 Crashes
Across multiple runs

~6,000 tok
VAPO response length

Ablation: Value-Pretraining Alone Contributes 49 Points

Model / Configuration	AIME 2024	Drop from VAPO	Score
Vanilla PPO	5	-55	
VAPO w/o Value-Pretraining	11	-49	
VAPO w/o Decoupled-GAE	33	-27	
VAPO w/o Length-Adaptive GAE	45	-15	
VAPO w/o Clip-Higher	46	-14	
VAPO w/o Token-level Loss	53	-7	
VAPO w/o Positive Example LM Loss	54	-6	
VAPO w/o Group-Sampling	55	-5	
VAPO (Full)	60	—	

All variants trained from Qwen2.5-32B with no SFT. Source: VAPO paper (arxiv 2504.05118)

Key Takeaways

VAPO: Value-Model-Based RL Is Back — With the Right Engineering

Value-Model RL Can Beat Value-Free

When bias, heterogeneous lengths, and reward sparsity are systematically addressed, value-model-based RL surpasses GRPO/DAPO

Value-Pretraining Is Non-Negotiable

Removing it drops AIME from 60 → 11. Without proper initialization, value-model RL barely functions.

Length-Adaptive GAE: A Novel Critical Technique

Dynamically adjusting λ per response length contributes 15 points. Unique to long-CoT reasoning settings.

SOTA in 5K Steps, Zero Crashes

VAPO achieves 60.4 on AIME 2024 within ~5,000 steps with no training instability across multiple runs.

Implications for Future Reasoning Models

The three-challenge framework (bias, heterogeneous lengths, reward sparsity) provides a principled lens for designing future value-model RL systems for long-CoT reasoning.